

b)	A	B	C	$A \cap B \cap C$	$A \cap B \cap C$	A	B	C	$A \cup B \cup C$
	1	1	1	1	0	0	0	0	0
	1	1	0	0	0	0	0	1	1
	1	0	1	0	0	0	1	0	1
	1	0	0	0	0	0	1	1	1
	0	1	1	0	0	1	0	0	1
	0	1	0	0	0	1	0	1	1
	0	0	1	0	0	1	1	0	1
	0	0	0	0	0	1	1	1	1

20) Let A, B and C be sets. Show that

a) $(A \cup B) \subseteq (A \cup B \cup C)$

Solution: Let, x be an arbitrary element of $A \cup B$, i.e. $x \in A \cup B$.
 By Defn. 1, $x \in A \vee x \in B$. By Defn. of OR, $(x \in A \vee x \in B) \vee (x \in C)$. By
 Defn. 1, $x \in A \cup B \cup C$. By Defn. 1, $x \in (A \cup B \cup C)$.
 By Defn. 3 from Section 2.1, $A \cup B \subseteq A \cup B \cup C$. $\square$

b) $(A \cap B \cap C) \subseteq (A \cap B)$

Solution: Let, x be an arbitrary element of $A \cap B \cap C$, i.e. $x \in A \cap B \cap C$.
 By Defn. 2, $x \in A \cap B \wedge x \in C$. By Defn. of AND, $x \in A \cap B$ is True.
 By Defn. 3 from Section 2.1, $A \cap B \cap C \subseteq A \cap B$. $\square$

c) $(A - B) - C \subseteq (A - C)$

Solution: Let, x be an arbitrary element of $(A - B) - C$, i.e. $x \in (A - B) - C$.
 By Defn. 4, $x \in (A - B) \wedge x \notin C$. By Defn. 4, $(x \in A \wedge x \notin B) \wedge x \notin C$. By 2nd
 Associative Law of propositions, $x \in A \wedge (x \notin B \wedge x \notin C)$. By 2nd Commutative Law
 of propositions, $x \in A \wedge (x \notin C \wedge x \notin B)$. By 2nd Associative Law of propositions,
 $(x \in A \wedge x \notin C) \wedge (x \notin B)$. By Defn. 4, $x \in (A - C) \wedge x \notin B$. By Defn. of AND, $x \in (A - C)$.
 By Defn. 3 from Sec-2.1, $(A - B) - C \subseteq (A - C)$. $\square$

d) $(A - C) \cap (C - B) = \emptyset$

Solution: $(A - C) \cap (C - B) = \{x \mid x \in (A - C) \wedge x \in (C - B)\}$ (By Defn. 2)
 $= \{x \mid (x \in A \wedge x \notin C) \wedge (x \in C \wedge x \notin B)\}$ (By Defn. 4)
 $= \{x \mid ((x \in A \wedge x \notin C) \wedge x \in C) \wedge x \notin B\}$ (By 2nd Associative
 Law of Propositions)